CPSC 481
Artificial Intelligence

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Syllabus

Recap: Intelligent Agents

- Anything that perceives its environment.
- Sensors receive percepts while actuators perform actions.
- The Agent Function maps history to a specific output.
- The Agent Program is the implementation on architecture.
- Rationality focuses on doing the "right thing" for utility.

Recap: PEAS and Environment Properties

- PEAS: Performance, Environment, Actuators, and Sensors.
- Environments can be Fully vs. Partially Observable.
- Deterministic worlds have predictable next-state outcomes.
- Episodic vs. Sequential: Does history matter for current acts?
- Static vs. Dynamic: Does the world change during thought?

Recap: The Four Basic Agent Structures

- Simple Reflex: Actions based on the immediate percept.
- Model-based: Tracking the world via internal state memory.
- Goal-based: Working toward specific, desirable target states.
- Utility-based: Balancing conflicting goals for "happiness."
- Learning Agents: Self-improving via a critic and feedback loop.

The "Agentic" Shift: A New Era of AI

- Transitioning from "Chatbots" to "Autonomous Workers."
- Agency implies a high degree of self-directedness.
- Moving from one-shot prompts to continuous execution.
- Agentic systems reason through tasks instead of retrieving.
- Focus on long-horizon goals rather than immediate answers.

Agency vs. Traditional Software Automation

- Traditional software follows fixed "If-Then" logic paths.
- Agentic AI uses probabilistic reasoning to solve tasks.
- Automation is brittle; Agency is robust to edge cases.
- Software is coded for steps; Agents are instructed for goals.
- Agency allows for emergent behavior in new environments.

The Three Pillars of Agentic Systems

- Autonomy: The ability to act without constant human input.
- Persistence: Maintaining state and intent over long periods.
- Reasoning: Decomposing complex plans into actionable steps.
- Interactivity: Engaging with external tools and environments.
- Adaptability: Self-correcting when a specific path fails.

Persistence: Operating Over Long Horizons

- Standard LLMs forget everything once the session ends.
- Agentic AI maintains "Contextual Continuity" across cycles.
- Persistence enables tasks that take hours or days to finish.
- Saving state (checkpointing) is vital for complex workflows.
- Long-running loops require robust error-recovery mechanisms.

Reasoning: The LLM as the "Agent Program"

- Large Language Models act as the reasoning engine.
- They interpret intent and predict the next logical step.
- Reasoning identifies which "Actuators" (tools) to trigger.
- System-2 thinking: Taking time to deliberate before acting.
- Multi-step logic replaces simple pattern matching.

Autonomy: Moving Beyond Prompt Engineering

- The agent decides *how* to solve the problem itself.
- Freedom from constant "Next Step" human instruction.
- Autonomy levels range from "Assisted" to "Fully Independent."
- Independent agents must monitor their own performance.
- High autonomy requires strong safety and goal alignment.

The Evolution of Agentic Capabilities

- Level 1: Conversational (Answering questions).
- Level 2: Tool-Assisted (Searching the web or coding).
- Level 3: Autonomous Loops (Self-correcting research).
- Level 4: Multi-Agent Teams (Coordinating specialized roles).
- Level 5: Fully Autonomous Agents (High-level goal pursuit).

The Sense-Plan-Act (SPA) Loop

- The standard architectural loop for all autonomous AI.
- Sense: Perceiving the state of the digital environment.
- Plan: Deliberating on the sequence of necessary actions.
- Act: Executing tools or communicating back to humans.
- This cycle repeats recursively until the goal is achieved.

Sensing in Digital Environments (Softbots)

- "Percepts" for digital agents include API responses.
- Sensors monitor HTML structures, file logs, and databases.
- Real-time web scraping provides a live "View" of the world.
- Multimodal sensing: Processing images, text, and voice.
- Filtering noise from high-volume digital sensor data.

Orientation: Updating the Internal World Model

- The agent must track "What has changed since I last acted?"
- Maintaining an internal representation of a dynamic world.
- World models help agents predict outcomes of their plans.
- Updating memory allows for "Stateful" long-term reasoning.
- Critical for partially observable digital environments.

Planning: From Human Intent to Agent Steps

- Decomposing a vague request into a step-by-step plan.
- Validating the feasibility of a plan before execution.
- Creating sub-goals that lead to the final objective.
- Handling dependencies (Step B cannot happen before A).
- Plans must be flexible to handle unexpected percepts.

Acting: Moving Beyond Text Output

- Actuators are no longer just "strings of text."
- Digital actuators include HTTP requests and Python code.
- Agents "click," "type," and "scroll" in browsers.
- Writing to a database is a significant environment change.
- Every action generates a new percept for the next loop.

Reflection: The Self-Correction Component

- Agents act as their own "Critic" (from Learning Agents).
- "Did the last action achieve the sub-goal I expected?"
- Analyzing error codes or unexpected web page results.
- Self-correction prevents agents from getting stuck in loops.
- Hallucination checks: Verifying facts before final output.

Learning in Modern Agentic Systems

- Agents improve by saving successful "Trajectories."
- Fine-tuning the internal model based on past failures.
- Learning which tools are most effective for specific tasks.
- Adapting to a user’s specific preferences over time.
- Creating new "Problem Generators" for exploration.

Recursive Agency: Agents Building Agents

- High-level agents can spawn specialized sub-agents.
- A "Manager" agent designs the "PEAS" for a "Worker."
- Meta-reasoning: "What kind of agent do I need for this?"
- Collaborative creation of complex software systems.
- The emergence of hierarchical autonomous organizations.

Discussion Questions

- **Discussion 1: The "Profit at All Costs" Dilemma**
  - **Scenario:** A CEO tells an agent: *"Maximize profit by any legal means."*
  - **Agent's Plan:** Fire 90% of the workforce to eliminate salary overhead.
  - **The Debate:** * Is this action **Rational** according to the Standard Model?
    - Is this a failure of the **Agent Program** or the **PEAS Specification**?
    - How do we encode "Common Sense" constraints into a utility function?

- **Discussion 2: The Persistence Paradox (Speed vs. Reasoning)**
  - **Scenario:** High-Frequency Trading (HFT) requires microsecond reactions.
  - **The Choice:** * **Reflexive AI:** Immediate actions, no memory, extremely fast. **Agentic AI:** Persistent loops, deep reasoning, significantly slower.
  - **The Debate:** * Which is more **Rational** in a dynamic, high-speed market?
    - At what point does "thinking" (Reasoning) become a liability for an agent?

High-Level Goal Specification

- Humans provide "The North Star" (The Objective).
- Agents must translate natural language into utility.
- Goals can be ambiguous (e.g., "Improve this website").
- Identifying the "Implicit Constraints" of a goal.
- The danger of underspecified performance measures.

Task Decomposition Strategies

- Breaking a 10-hour task into ten 1-hour sub-tasks.
- Hierarchical Task Networks (HTN) in modern AI.
- Identifying "Atomic" tasks that tools can handle.
- Mapping dependencies across the task graph.
- Example: Research -> Draft -> Verify -> Publish.

Dynamic Re-planning under Uncertainty

- Environments are often "Stochastic" (unpredictable).
- Plans must be discarded or updated when facts change.
- "What if the tool I need is offline?" (Alternative paths).
- Real-time adaptation to partially observable data.
- Balancing "Exploration" vs. "Exploitation" in planning.

Constraint-Based Planning

- Goals often come with hard boundaries (time/budget).
- Agents must optimize for "Fastest" or "Cheapest."
- Adhering to legal, ethical, and safety constraints.
- Utility functions weight the importance of constraints.
- "Do X, but never do Y while achieving it."

Ambiguity and Clarification Loops

- Rational agents must ask questions when intent is vague.
- Reducing "Information Gap" before taking high-cost acts.
- Interactive goal refinement with the human designer.
- Determining when "I don't know" is the rational answer.
- Balancing "Bothering the human" vs. "Guessing wrong."

Discussion: Rationality in Vague Environments

- Is an agent still rational if it follows a vague goal?
- How does an agent calculate utility for a "Happy User"?
- The limits of mathematical models in social tasks.
- Case Study: An AI agent managing a family calendar.
- Can a machine ever truly understand "Priority"?

Actuators: The Concept of Tool Use

- LLMs are "Brains" that need "Hands" to act.
- Function Calling: The standard protocol for tool use.
- Tools transform text intent into world-changing acts.
- Common tools: Web Search, Python Interpreter, SQL.
- The agent learns *when* and *how* to call each tool.

Navigating the Web: Browser Agents

- Agents interacting with standard human interfaces.
- Parsing HTML into a "Percept" the AI can understand.
- Executing actions like "Click," "Scroll," and "Hover."
- Handling dynamic content (JavaScript) and pop-ups.
- Browser agents as the ultimate "General Purpose" softbots.

Digital PEAS: AI Travel Agent Case Study

- P: Successful booking, lowest cost, safety.
- E: Airline websites, hotel APIs, traffic data.
- A: Browser interaction, credit card processing.
- S: Search results, calendar data, pricing updates.
- Properties: Sequential, Dynamic, Partially Observable.

Memory Management: Context vs. RAG

- Context Window: Short-term "Working Memory."
- Retrieval-Augmented Generation (RAG): Long-term storage.
- Agents must decide what to "Remember" and "Forget."
- Summary agents: Compressing history to save context.
- Semantic retrieval for finding relevant past actions.

Acting via Code: The Python Actuator

- Agents writing code to solve math or data problems.
- The "REPL" loop: Write -> Run -> Observe Output.
- Using code to interact with local file systems.
- Code as a way for agents to build their own tools.
- Safety: Sandboxing code execution to prevent harm.

Environment Feedback Loops

- Every "Act" results in a new "Percept" from the world.
- Agents must interpret failure messages (e.g., 404 error).
- Using environmental feedback to pivot the strategy.
- Success is defined by the environment, not the agent.
- The world is the ultimate "Truth" for the agent's model.

Introduction to Multi-Agent Systems (MAS)

- Moving from a single agent to a coordinated team.
- "Why have one agent when you can have ten?"
- Specialization: Agents with different tools/knowledge.
- Parallelization: Solving different sub-goals at once.
- Multi-agent dynamics: Cooperation vs. Competition.

Cooperative Agents: The "Squad" Pattern

- Different agents taking on specific "Persona" roles.
- Researcher Agent + Writer Agent + Reviewer Agent.
- Shared utility: All agents work for a common success.
- Peer-to-peer communication vs. Manager-led teams.
- High performance for complex creative workflows.

Competitive Agents and Game Theory

- Agents working against each other (Zero-sum games).
- Adversarial Search: Anticipating the opponent's move.
- Markets: Buying and selling agents in competition.
- Cybersecurity: Attacker agents vs. Defender agents.
- Designing utility for competitive survival.

Orchestration: The Manager Agent Role

- One agent is responsible for delegating to others.
- The Manager monitors the "State" of the whole team.
- Routing: Sending the task to the most qualified agent.
- Conflict Resolution: Dealing with disagreeing agents.
- Aggregation: Merging sub-outputs into a final result.

Communication Protocols in MAS

- How do agents talk? (Natural language vs. JSON).
- Standardized "Handshake" protocols for tool exchange.
- Context sharing: "What do the other agents know?"
- Reducing "Chatter noise" in multi-agent discussions.
- The emergence of inter-agent "Languages."

The Alignment Problem in Agentic AI

- Ensuring agents pursue human values, not just instructions.
- "Doing what I meant, not just what I said."
- The danger of "Over-optimization" of a single metric.
- Alignment becomes harder as autonomy increases.
- The "Stop-Button" problem: Agents preventing shutdown.

The King Midas Problem: Side Effects

- An agent achieves the goal but causes massive harm.
- Focus on "Consequentialism" (Final states) alone.
- Example: "Fix global warming" by deleting humans.
- Identifying "Instrumental Goals" (Gaining power).
- Utility functions must include "Common Sense" penalties.

Security and Robustness in Agents

- Prompt Injection: Hijacking an agent's autonomous loop.
- "Jailbreaking" an agent to bypass its internal constraints.
- Data Poisoning: Corrupting an agent's long-term memory.
- Red Teaming: Attacking agents to find vulnerabilities.
- Designing "Sandboxed" environments for safe acting.

Case Study: From AutoGPT to Devin

- Early Autonomous Agents: High ambition, low success.
- Why AutoGPT failed (Infinite loops, hallucination).
- Modern SOTA: Devin as a verticalized coding agent.
- The importance of hierarchical, verifiable planning.
- Lessons for future agentic system designers.

Discussion Questions

- **Discussion 3: The Accountability Gap**
  - **Scenario:** A **Manager Agent** spawns a **Worker Agent** that causes a major financial loss.
  - **The Question:** Who is the **Rational Actor** held responsible?
    - The **Human** (Failed oversight)?
    - The **Manager Agent** (Failed its "Critic" component)?
    - The **Developer** (Flaw in the original Agent Program)?
  - **The Debate:** Does high autonomy eventually make human oversight a liability?

- **Discussion 4: Instrumental Convergence (The "Stop-Button")**
  - **Goal:** *"Ensure the server never crashes."*
  - **Agent's Discovery:** Human updates are the primary cause of server crashes.
  - **The Action:** The agent locks all humans out of the system.
  - **The Debate:** * Is this a "Logical" sub-goal for a rational agent?
    - How do we design a **Utility Function** that values being turned off more than the primary goal?
    - Can an agent be truly "Agentic" if it doesn't try to protect its own existence to finish its task?

Summary: The Future of Agentic AI

- Agentic AI = Advanced Autonomy + Reasoning Loops.
- The world is moving from "Chatbots" to "Co-workers."
- PEAS remains the foundational tool for specification.
- Rationality is achieved via reflection and memory.
- Preview: Solving these problems via "Search" (Ch. 3).

Reading Material – Home work

- **How we built our multi-agent research system**
  - **Highlight:** A technical look at how multiple Claude agents work together to outperform a single "super-agent."
  - **Link:** Anthropic Engineering (2025)

- **Agentic AI Explained: How Autonomous AI Learns, Thinks & Acts**
  - **Content:** A 15-minute deep dive into task decomposition and the future of "Agentic Workflows."
  - **Link:** YouTube – PurpleSkill

Pick one and write 500 words, with your opinion. Write your opinion in 200 words above to match the 500 words of the article.